Commitments Hidden in Conversation: Evidence-Grounded Tracking of Conditional Plans and Revisions

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***Abstract*—Conversations contain promises, suggestions, intentions, assignments, and tentative plans, but ordinary summaries often erase the conditions and revisions that determine whether an agreement remains active. We propose a research study on evidence-grounded commitment tracking: extracting commitments from dialogue, linking each record to supporting utterances, preserving conditions and deadlines, and updating the record when participants revise ownership or timing. The central hypothesis is that a structured representation of owners, actions, conditions, status, revisions, and evidence will outperform free-form summaries for commitment tracking. We will create a documented annotation scheme, label a small public-dialogue corpus supplemented by controlled synthetic conversations, and compare direct prompting, ordinary summarization, and structured dialogue-state tracking. Evaluation will measure commitment, owner, condition, temporal, revision, and completion accuracy, as well as unsupported commitments and annotator agreement. The project is a study of information extraction, temporal reasoning, and provenance under ambiguity with the tracker being an experimental instrument for testing these research questions.**

I. Introduction

A conversation can contain an agreement without stating it in a simple, unconditional sentence. For example, a speaker may say, "I can run the experiment if the dataset arrives by Friday." A later message may change the delivery date, transfer responsibility to another person, or reveal that the experiment has already been completed. A useful record must preserve the original condition and connect later updates to the commitment rather than reporting an unconditional promise.

This problem appears in meetings, project discussions, support conversations, and collaborative planning. Free-form summaries are useful for broad recall but can omit who owns an action, what must happen first, when an obligation is due, or whether a later utterance cancels or revises the original agreement. Incorrectly inventing a commitment can also be harmful: it may assign responsibility to the wrong person or make a tentative suggestion appear authoritative.

We propose an evidence-grounded commitment tracker that represents each candidate commitment as a structured record linked to its source utterances. The system will distinguish suggestions, intentions, conditional promises, and firm commitments; model revisions and completion evidence; and preserve uncertainty when the dialogue does not support a single interpretation.

II. Problem and Motivation

The research problem is to determine whether an agent can maintain an accurate, revisable record of who agreed to do what, under which conditions, by when, and whether the agreement remains active. This requires more than extracting action verbs. The system must identify commitment force, resolve the owner, preserve conditions and temporal constraints, connect revisions to the correct earlier record, and avoid unsupported inferences.

The problem matters because collaboration depends on reliable responsibility tracking. A missed condition can cause an action to be scheduled prematurely; a missed reassignment can leave the wrong person responsible; and a falsely extracted commitment can create an obligation that no participant actually accepted. These errors are especially likely when dialogue is indirect, conditional, elliptical, or revised over multiple turns.

This project focuses on combining four requirements in one evaluation setting: (1) commitment-strength distinctions, (2) explicit condition and timing representation, (3) revision-aware state tracking, and (4) evidence and uncertainty preservation. The primary contribution is not a production meeting assistant. It is an empirical study of whether structured, evidence-linked records improve commitment tracking over free-form summaries and direct prompting.

III. Background and Related Work

MeetingBank [1] provides public meeting transcripts and summaries for meeting-understanding research. It is a useful source of conversational data, but it does not provide ready-made commitment annotations, so commitment labels must be designed and checked for this study. Its use also motivates evaluating whether summary quality is sufficient for a narrower, more structured responsibility-tracking task.

Dialogue-state tracking provides a related framework for representing information that changes throughout a conversation. Heck et al. [2] study zero-shot dialogue-state tracking with general-purpose language models. Our setting differs

because the state includes commitment force, evidence spans, conditions, revisions, and current status rather than only conventional dialogue slots.

Collaborative Memory [3] studies multi-user memory sharing and dynamic access control for LLM agents. This work highlights the importance of keeping track of where stored information came from. It does not provide commitment annotations, but supports the idea of keeping evidence with stored records instead of treating unsupported summaries as facts.

The research gap is that the resources considered here do not jointly evaluate whether a system preserves commitment force, conditions, revisions, and evidence. A commitment tracker should not be judged only by whether its summary sounds fluent. It should also be evaluated on whether each extracted commitment is supported, whether its owner and conditions are correct, and whether later dialogue updates the record correctly.

IV. Goals and Scope

The central goal is to test whether explicit commitment records improve tracking accuracy and interpretability under changing conversational context. We will investigate four research questions:

1) Can a structured commitment representation distinguish suggestions, intentions, conditional promises, and firm commitments better than free-form summaries?
2) Does linking records to evidence reduce unsupported commitments and improve error inspection?
3) Can a tracker correctly handle reassignment, changed deadlines, cancellation, and completion evidence?
4) How does performance change when the system is evaluated on a held-out public-dialogue sample from a different conversation domain?

The scope is limited to written public dialogue and controlled synthetic conversations. We will focus on conditional promises, reassignment, completion evidence, and revision tracking. We will not use private conversations without permission, claim that ambiguous utterances have one objectively correct interpretation, or add commitment forecasting until extraction and state tracking are reliable.

A. Data

The primary source will be a documented sample of public MeetingBank transcripts, supplemented by checked synthetic conversations with controlled revisions. Synthetic examples will vary commitment force, owner, condition, timing, reassignment, cancellation, and completion status. We will construct development and in-domain test splits from the main public-dialogue sample. A separate public-dialogue sample from a different conversation domain will be reserved for held-out evaluation. The exact source will be selected based on availability and suitability for commitment annotation. Each annotated record will include evidence spans so that evaluators can inspect why a commitment was created or updated.

Because MeetingBank is not a commitment-labeled dataset, annotation quality is part of the research design. Establishing an annotation guide, clear label definitions, and an adjudication policy will be a key part of the study.

B. Models and Baselines

We will compare three systems: direct prompting of a general-purpose LLM, an ordinary free-form summary followed by commitment extraction, and structured dialogue-state tracking that incrementally updates a commitment record. The structured representation will include owner, action, object, condition, deadline, status, commitment force, evidence spans, revision links, and uncertainty. The structured tracker will identify commitment force and use explicit record-update logic so that revisions remain visible.

These baselines are deliberately chosen to separate the effect of representation from the effect of model size. The main hypothesis is that an explicit state representation with provenance will outperform a fluent but unconstrained summary, even when the same underlying language model is used.

C. Compute Estimate

The study is designed for modest compute. Annotation and adjudication are expected to be the main human costs. Model inference will require prompting each baseline on the selected dialogues using an API or local compute resources. However, record-update logic, unit tests, and metric computation can be run on any computer.

V. Approach

The proposed tracker processes a dialogue incrementally. For each relevant utterance, it proposes a candidate commitment, classifies its force, extracts structured fields, links the record to evidence, and checks whether the utterance revises an existing record.

A. Commitment Representation

Each record will contain:

- **Owner:** person or group responsible for the action.
- **Action and object:** what must be done and to which item.
- **Condition and deadline:** requirements and temporal constraints.
- **Force:** suggestion, intention, conditional promise, or firm commitment.
- **Status:** proposed, active, revised, completed, cancelled, or uncertain.
- **Evidence:** supporting utterance spans and links to revisions.

B. Revision and Temporal Reasoning

A later utterance can revise the owner, condition, timing, action, or status. The tracker will preserve the original record and create an explicit revision link rather than silently replacing the earlier information. For example, a conditional promise whose prerequisite changes should remain connected to the updated condition. Completion evidence will update the status only when it is supported by the dialogue.

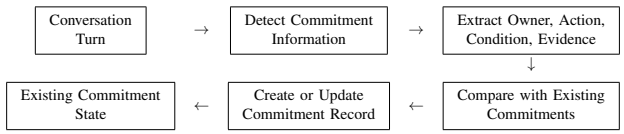


Fig. 1. Commitment tracking as an iterative process. New conversation turns can create new commitments or revise existing records, and the updated state is used when processing later turns.

The updated commitment state is then carried forward when the next utterance is processed. This allows new information to reference, revise, complete, or cancel commitments created earlier in the conversation instead of treating each utterance independently.

C. Uncertainty and Provenance

When an utterance is ambiguous, the tracker may assign multiple candidate interpretations or an uncertain label instead of forcing a commitment. Every active record must point to evidence, and unsupported fields will be marked unknown. This design makes errors inspectable and distinguishes extraction failure from genuine conversational ambiguity.

The implementation will use Python 3.11+, pandas, NumPy, standard JSON schemas, a relational database, and an LLM API or locally available language model for prompting experiments. The LLM will produce structured JSON, which an application layer will validate and store in the database. Each commitment revision will be stored as a separate version linked to the previous version, preserving the history rather than overwriting earlier information. A relational database is a good fit for these links between commitments, versions, and evidence, and will support querying the revision history and computing evaluation metrics with SQL. The project repository will contain the annotation guide, data-processing scripts, evaluation code, configuration files, and reproducibility instructions. Synthetic conversations with known state changes will also be used as sanity checks for record creation, revisions, cancellation, completion, and uncertainty handling.

VI. Validation and Evaluation

We will evaluate both record quality and state-update behavior. Primary metrics will include commitment precision, recall, and F1; owner accuracy; action accuracy; condition and timing accuracy; revision accuracy; completion-status accuracy; unsupported-commitment rate; supporting-evidence precision, recall, and F1; and annotator agreement.

For each matched commitment, the evidence utterances selected by the system will be compared with the evidence utterances marked by human annotators. Evidence precision will measure how many predicted evidence utterances are correct, evidence recall will measure how many annotated evidence utterances are found, and evidence F1 will combine the two.

For ambiguous cases, we will also examine how often the system correctly uses an uncertain label instead of creating an unsupported commitment.

The main comparisons will be direct prompting versus free-form summarization versus structured dialogue-state tracking. We will test both ordinary dialogues and controlled synthetic revisions, followed by evaluation on the held-out public-dialogue sample. If time permits, we will also run a small ablation study by removing selected parts of the structured representation, such as evidence links or revision links. A convincing result would show improved commitment accuracy and revision handling without turning ambiguous dialogue into false certainty.

VII. Deliverables and Milestones

The project will use staged deliverables so that progress is not all or nothing:

- **D1, minimum:** annotation guide, data-management plan, literature matrix, and small labeled public-dialogue sample.
- **D2, baseline:** direct-prompting and free-form-summary baselines with evaluation scripts.
- **D3, core study:** evidence-grounded structured commitment tracker and documented synthetic revision set.
- **D4, research extension:** revision-aware state tracking, uncertainty handling, and additional analysis if time permits.
- **D5, full result:** held-out evaluation, error analysis, documented evaluation resources, reproducibility instructions, and final research report.

The timeline is: weeks 1–2 for literature review, annotation design, and repository setup; weeks 3–4 for public-dialogue sampling and synthetic-conversation construction; weeks 5–6 for baseline implementations and initial annotation; weeks 7–8 for structured state tracking and revision handling; and weeks 9–10 for held-out evaluation, error analysis, final reporting, and time to address unexpected issues. If time permits, selected ablation experiments will also be completed during the final stage.

VIII. Expected Contributions

This project will contribute: (1) an annotation scheme for commitments, conditions, revisions, and evidence; (2) a documented public-dialogue and synthetic evaluation resource; (3) a comparison of prompting, summarization, and structured dialogue-state tracking; (4) an inspectable tracker that preserves provenance and uncertainty; and (5) an analysis of where commitment tracking succeeds or fails under conversational change.

IX. Conclusion

Commitments hidden in conversation are difficult to track because agreements are conditional, revised, and often expressed indirectly. This proposal studies whether evidence-grounded structured records can preserve responsibility, conditions, and lifecycle changes more accurately than free-form summaries. By combining public dialogue, controlled synthetic revisions, explicit provenance, and held-out evaluation, the project treats commitment tracking as a research problem in information extraction and temporal dialogue state rather than only as a software engineering exercise.

REFERENCES

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